3

Direct Methods for Solving A System of Linear Equations

Chapter Learning Outcomes

CLO

3

- Interpret the meaning of the solution of a system of linear equations.
- Express a system of linear equations in a matrix form.
- Write the augmented matrix for a system of linear equations.
- Know some methods for solving a system of linear equations.
- Compute the determinant of a square matrix.
- Apply Cramer's method for solving a system of linear equations.
- Do the elementary row operations for the matrices and systems of linear equations.
- Investigate the solution of a system using Gauss elimination method.
- Investigate the solution of a system using Gauss-Jordan elimination method.
- Identify inconsistent (parallel) systems of equations containing two variables.
- Identify dependent (coincide) systems of equations containing two variables.
- Find the inverse of a given matrix.
- Apply the inverse method for solving a system of linear equations.

Direct Methods for Solving a System of Linear Equations

An equation is said to be **linear** if every variable has degree equal to one (or zero).

EXAMPLE

$$4x+5y-2z=8$$

Linear Equation

$$2x^{3}+3y^{-1}+\sqrt{z}=8$$

Not a Linear Equation

Direct Methods for Solving a System of Linear Equations

- A **system of linear equations** is a group of two or more linear equations that use the same variable.
- To solve a system of equations means to find values for the variables that satisfy all of the equations in the system.

EXAMPLE

$x = 5$ $y = 3$

$$2x + 4y = 22$$
$$x - 6y = -13$$

$$2 \cdot 5 + 4 \cdot 3 = 22$$
$$5 - 6 \cdot 3 = -13$$

The ordered pair (5, 3) is a solution of the linear equations below.

Standard and Matrix Forms

Standard Form

$$
\begin{aligned}
2x + 3y + 5z &= 8 \\
x + y + z &= 5 \\
4x + 5z &= 2
\end{aligned}
$$

Matrix Form

Coefficient Matrix

Constant Matrix

$$
\underbrace{\begin{pmatrix} 2 & 3 & 5 \\ 1 & 1 & 1 \\ 4 & 0 & 5 \end{pmatrix}}_{A}
\underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{X}
=
\underbrace{\begin{pmatrix} 8 \\ 5 \\ 2 \end{pmatrix}}_{B}
$$

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Augmented Matrix

Standard Form

$$
\begin{aligned}
2x + 3y + 5z &= 8 \\
x + y + z &= 5 \\
4x + 5z &= 2
\end{aligned}
$$

Augmented Matrix

$$
\underbrace{\begin{pmatrix} 2 & 3 & 5 \\ 1 & 1 & 1 \\ 4 & 0 & 5 \end{pmatrix}}_{A}
\left|
\underbrace{\begin{pmatrix} 8 \\ 5 \\ 2 \end{pmatrix}}_{B}
\right.
$$

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Quiz

For the following system of linear equations:

1- Convert it to a matrix form.

2- Write the augmented matrix.

$$\begin{aligned} 3x - y + z &= 3 \\ 2x - 4y + 3z &= 16 \\ x - y + z &= 5 \end{aligned}$$

Matrix Form

Augmented Matrix

Matrix Determinant (2 × 2)

- Given a **square matrix** A, its **determinant** is a real number associated with the matrix.
- The determinant of A is written:

$$\det(A) \quad \text{or} \quad |A|$$

- For a 2x2 matrix, the definition is:

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Matrix Determinant (2 × 2)

Multiply

$$\begin{pmatrix} -4 & 8 \\ 5 & -6 \end{pmatrix}$$

Multiply

40 (2nd)

24 (1st)

Now subtract these two numbers.

24 (1st) - 40 (2nd) = **-16**

-16 is the **determinant** of this matrix

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Matrix Determinant (2 × 2)

EXAMPLE

Find the determinant for the following matrix:

$$A = \begin{pmatrix} -5 & 2 \\ -2 & 0 \end{pmatrix}$$

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Matrix Determinant (3 × 3)

- For a 3x3 matrix, the determinant is as follows:

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \qquad \left(\begin{array}{ccc|cc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{array}\right)$$

$$|A| = [aei + bfg + cdh] - [gec + hfa + idb]$$

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Matrix Determinant (3 × 3)

$$\det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} =$$

(gec +hfa +idb)

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{matrix} a & b \\ d & e \\ g & h \end{matrix}$$

(aei+bfg +cdh)

Now Subtract the 2nd set products from the 1st.

(aei + bfg + cdh) - (gec + hfa + idb)

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Matrix Determinant (3 × 3)

EXAMPLE

Find the determinant for the following matrix:

$$A = \begin{pmatrix} 2 & 0 & 3 \\ 1 & -2 & 5 \\ -1 & 4 & 2 \end{pmatrix}$$

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Quiz

Find the determinant for the following matrix:

$$A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 3 \\ 2 & 7 & 0 \end{pmatrix}$$

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Quiz

$$\det\begin{pmatrix} 2 & -1 & 3 \\ -2 & 0 & 1 \\ 1 & 2 & 4 \end{pmatrix}$$

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Methods for Solving a System of Linear Equations

Direct Methods	Iterative Methods
Cramer's	Jacobi
Gauss Elimination	Gauss-Seidel
Gauss-Jordan Elimination	
Inverse	

Cramer's Method

Cramer's Method

In general the solution to the system

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases}$$

is (x,y)

where

$$x = \frac{\begin{vmatrix} e & b \\ f & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

$$y = \frac{\begin{vmatrix} a & e \\ c & f \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

and $\begin{vmatrix} a & b \\ c & d \end{vmatrix} \neq 0$

If we let **A** be the **coefficient matrix** of the linear system, notice this is just ***det A***.

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Cramer's Method

In general the solution to the system

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases}$$

$$x = \frac{|A_x|}{|A|}$$

$$y = \frac{|A_y|}{|A|}$$

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Cramer's Method

EXAMPLE

Use Cramer's method to solve the following system of linear equations.

$$\begin{cases} 2x - 3y = 12 \\ -x + 2y = 9 \end{cases}$$

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Cramer's Method - Steps

$$\begin{cases} 2x - 3y = 12 \\ -x + 2y = 9 \end{cases}$$

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Cramer's Method - Steps

$$\begin{cases} 2x - 3y = 12 \\ -x + 2y = 9 \end{cases}$$

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Cramer's Method - Steps

$$\begin{cases} 2x - 3y = 12 \\ -x + 2y = 9 \end{cases}$$

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Cramer's Method - Steps

$$\begin{cases} 2x - 3y = 12 \\ -x + 2y = 9 \end{cases}$$

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Cramer's Method

EXAMPLE

Use Cramer's method to solve the following system of linear equations.

$$4x + 2y = 10$$
$$5x + y = 17$$

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Cramer’s Method

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Cramer's Method

EXAMPLE

Use Cramer's method to solve the following system of linear equations.

$$6x + 4y = 10$$

$$3x + 2y = 5$$

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Cramer’s Method

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Cramer's Method: (3x3) Matrix

Let A be the coefficient matrix of this linear system:

$$\begin{cases} ax+by+cz=j \\ dx+ey+fz=k \\ gx+hy+iz=l \end{cases} \Rightarrow A=\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

If **det A** is **not 0**, then the system has **exactly one solution**. The solution is:

$$x=\frac{\begin{pmatrix} j & b & c \\ k & e & f \\ l & h & i \end{pmatrix}}{\det A} \qquad y=\frac{\begin{pmatrix} a & j & c \\ d & k & f \\ g & l & i \end{pmatrix}}{\det A} \qquad z=\frac{\begin{pmatrix} a & b & j \\ b & e & k \\ c & h & l \end{pmatrix}}{\det A}$$

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Cramer's Method

EXAMPLE

Use Cramer's method to solve the following system of linear equations.

$$2x + 3y + 5z = 8$$
$$x + y + z = 5$$
$$4x + 5z = 2$$

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Cramer's Method: (3x3) Matrix

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Cramer's Method: (3x3) Matrix

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Cramer's Method: (3x3) Matrix

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Cramer’s Method

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Elementary Row Operations

Elementary Row Operations (1)

- **Swap two rows**

$$\xrightarrow{R1 \leftrightarrow R2} \left(\begin{array}{ccc|c} 2 & 3 & 5 & 8 \\ 1 & 1 & 1 & 5 \\ 4 & 0 & 5 & 2 \end{array}\right)$$

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Elementary Row Operations (2)

- Multiply one of the rows by a nonzero constant.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 13 & -26 \end{array}\right) \xrightarrow{R3 = \frac{1}{13} R3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 1 & -2 \end{array}\right)$$

Elementary Row Operations (3)

- Add a multiple of one row to another row.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 4 & 0 & 5 & 2 \end{array}\right) \xrightarrow{R3 = -4R1+R3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & -4 & 1 & -18 \end{array}\right)$$

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Elementary Row Operations (Summary)

- Swap two rows
- Multiply one of the rows by a nonzero constant.
- Add a multiple of one row to another row.

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Methods for Solving a System of Linear Equations

Direct Methods	Iterative Methods
Cramer's	Jacobi
Gauss Elimination	Gauss-Seidel
Gauss-Jordan Elimination	
Inverse	

Gauss Elimination Method

Gauss Elimination Method

1- It is a direct method to solve a system of linear equations.

2- It based on converting the augmented matrix to row echelon form (REF).

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Gauss Elimination Method

Convert the augmented matrix to row echelon form (REF)

Augmented Matrix

$$\left(\begin{array}{ccc|c} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{array}\right) \longrightarrow$$

Row Echelon Form

$$\left(\begin{array}{ccc|c} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{array}\right)$$

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Gauss Elimination Method

EXAMPLE

Solve the following system of linear equations using Gauss elimination method.

$$2x + 3y = -1$$
$$3x + 2y = 1$$

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Gauss Elimination Method

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Gauss Elimination Method

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Gauss Elimination Method

EXAMPLE

Solve the following system of linear equations using Gauss elimination method.

$$
\begin{aligned}
2x + 3y + 5z &= 8 \\
x + y + z &= 5 \\
4x + 5z &= 2
\end{aligned}
$$

X = 3 Y= 4 Z = -2

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Gauss Elimination Method

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Methods for Solving a System of Linear Equations

Direct Methods	Iterative Methods
Cramer's	Jacobi
Gauss Elimination	Gauss-Seidel
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Inverse	

Gauss-Jordan Elimination Method

Gauss-Jordan Elimination Method

1- It is a direct method to solve a system of linear equations.

2- It based on converting the augmented matrix to reduced row echelon form (RREF).

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Gauss-Jordan Elimination Method

Convert the augmented matrix to reduced row echelon form (RREF)

Augmented Matrix

$$\left(\begin{array}{ccc|c} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{array}\right) \longrightarrow$$

RREF

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array}\right)$$

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Reduced Row Echelon Form (RREF)

DEFINITION Reduced Form

A matrix is said to be in **reduced row echelon form**, or, more simply, in **reduced form**, if

1. Each row consisting entirely of zeros is below any row having at least one nonzero element.
2. The leftmost nonzero element in each row is 1.
3. All other elements in the column containing the leftmost 1 of a given row are zeros.
4. The leftmost 1 in any row is to the right of the leftmost 1 in the row above.

Reduced Row Echelon Form (RREF)

The following matrices are in reduced form. Check each one carefully to convince yourself that the conditions in the definition are met.

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -3 \end{array}\right] \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right] \quad \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & 4 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 6 \end{array}\right] \quad \left[\begin{array}{ccc|c} 1 & 0 & 4 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Gauss-Jordan Elimination Method

Augmented Matrix

$$\left(\begin{array}{ccc|c} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{array}\right)$$

Gauss

REF

$$\left(\begin{array}{ccc|c} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{array}\right)$$

Gauss-Jordan

RREF

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array}\right)$$

Gauss-Jordan Elimination Method

EXAMPLE

Solve the following system of linear equations using Gauss-Jordan elimination method.

$$
\begin{aligned}
y - 8z &= 9 \\
x - 2y + 3z &= -3 \\
7y - 5z &= 12
\end{aligned}
$$

X = 2 **Y= 1** **Z = -1**

Gauss-Jordan Elimination Method

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Gauss-Jordan Elimination Method

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Gauss-Jordan Elimination Method

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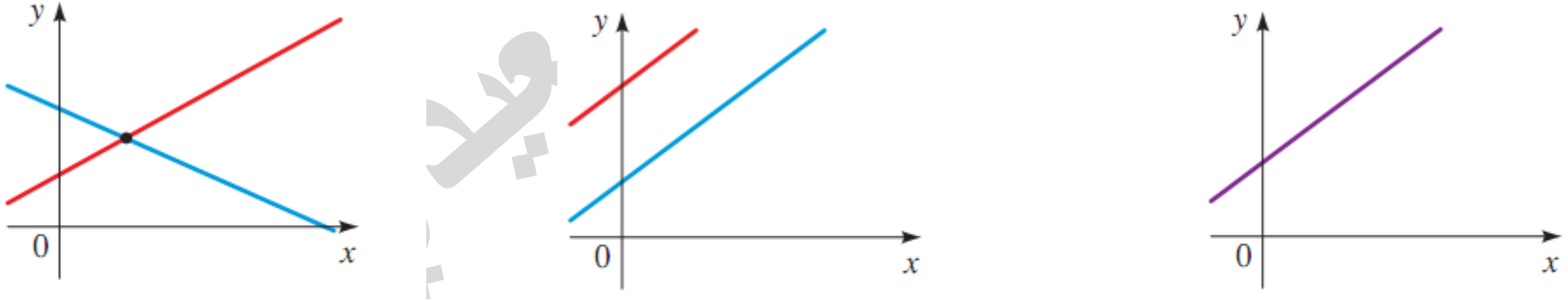
Three - Cases

Direct Methods for Solving Linear System of Equations

NUMBER OF SOLUTIONS OF A LINEAR SYSTEM IN TWO VARIABLES

For a system of linear equations in two variables, exactly one of the following is true. (See Figure 5.)

1. The system has exactly one solution.
2. The system has no solution.
3. The system has infinitely many solutions.



(a) Lines intersect at a single point. The system has one solution.

(b) Lines are parallel and do not intersect. The system has no solution.

(c) Lines coincide—equations are for the same line. The system has infinitely many solutions.

Matrix Representations of Consistent, Inconsistent and Dependent Systems

- The following matrix representations of three linear equations in three unknowns illustrate the three different cases:

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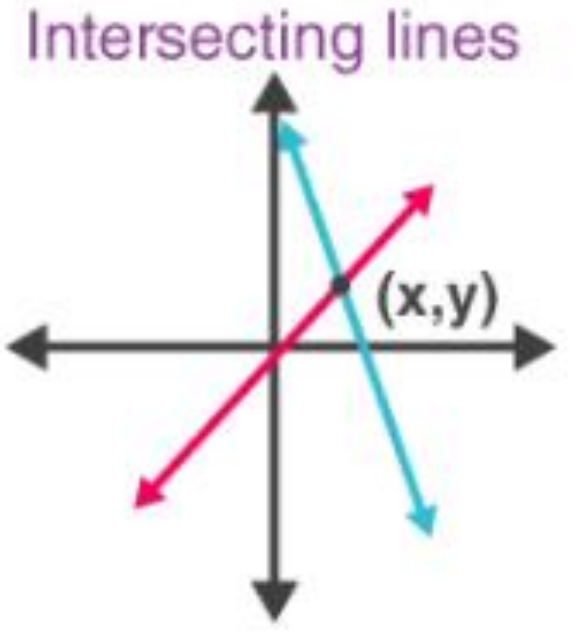
Case 1: The System has Exactly One Solution

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right)$$

- From this matrix representation, you can determine that

$x = 2$
$y = 1$
$z = -1$

Consistent System



One point in common
Solution ; (x,y)

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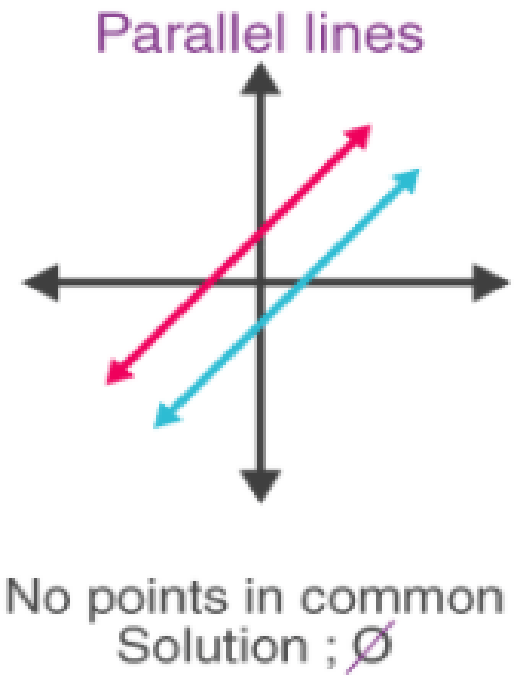
Case 2: The System has No Solution

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & -4 & 1 \\ 0 & 0 & 0 & -1 \end{array}\right)$$

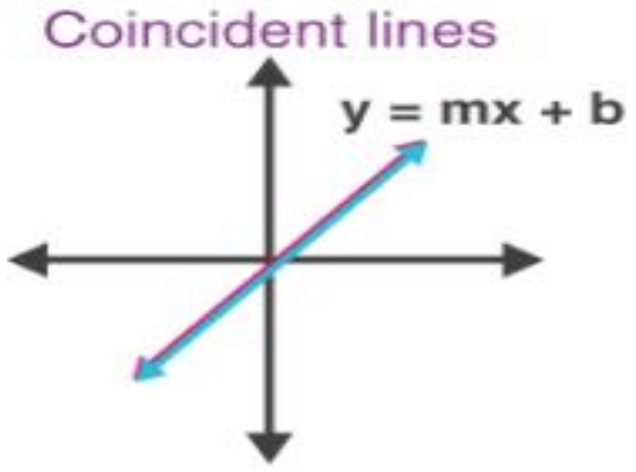


NOTE

- If there is a row in the augmented matrix containing all zeros to the left of the vertical line and nonzero entry to the right of the line, then the system of equations has no solution.
- The system of equations is inconsistent (parallel).

Case 3: The System has Infinitely Many Solutions

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$



Infinitely many points in common;
{(x,y) : y= mx + b}

NOTE

- If there is a row in the augmented matrix containing all zeros to the left of the vertical line and zero entry to the right of the line, then the system of equations has infinite number of solutions.
- The system of equations is dependent (coincide).

Gauss-Jordan Elimination Method

EXAMPLE

Solve the following system of linear equations using Gauss-Jordan elimination method and determine the case of this system.

$$
\begin{aligned}
x + 2y - 3z &= -2 \\
3x - y - 2z &= 1 \\
2x + 7y - 5z &= -3
\end{aligned}
$$

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Gauss-Jordan Elimination Method

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Gauss-Jordan Elimination Method

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Gauss-Jordan Elimination Method

EXAMPLE

Solve the following system of linear equations using Gauss-Jordan elimination method and determine the case of this system.

$$3x + 6y - 9z = 15$$

$$2x + 4y - 6z = 10$$

$$-2x - 3y + 4z = -6$$

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Gauss-Jordan Elimination Method

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Gauss-Jordan Elimination Method

Solve the following system of linear equations using Gauss-Jordan elimination method and determine the case of this system.

$$2x - 4y + z = -4$$

$$4x - 8y + 7z = 2$$

$$-2x + 4y - 3z = 5$$

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Gauss-Jordan Elimination Method

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Gauss-Jordan Elimination Method

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Summary of the 3-Cases

The linear system
$$\begin{cases} ax + by + cz = j \\ dx + ey + fz = k \\ gx + hy + iz = l \end{cases}$$

must have

Case	No. of Solutions	System Type	
1	Exactly one solution	Consistent and Independent	1 0 0 | *
2	No solution	Inconsistent (Parallel)	0 0 0 | *
3	Infinitely many solutions	Consistent and dependent (coincide)	0 0 0 | 0

Methods for Solving a System of Linear Equations

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Inverse	

Inverse Method

Inverse Method

If $AX = B$ is a linear system of n equations in n unknowns and if A^{-1} exists, then

$$X = A^{-1}B$$

is the unique solution of the system.

Inverse Method

$$
\begin{aligned}
2x + 3y + 5z &= 8 \\
x + y + z &= 5 \\
4x + 5z &= 2
\end{aligned}
$$

$$A.X = B$$

$$X = A^{-1}B$$

Coefficient Matrix

Constant Matrix

$$\underbrace{\begin{pmatrix} 2 & 3 & 5 \\ 1 & 1 & 1 \\ 4 & 0 & 5 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{X} = \underbrace{\begin{pmatrix} 8 \\ 5 \\ 2 \end{pmatrix}}_{B}$$

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Inverse Method

EXAMPLE

Solve the following system of linear equations using the inverse method.

$$2x + y + z = 1$$
$$3x + 2y + z = 2$$
$$2x + y + 2z = -1$$

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Inverse Method

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Inverse Method

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How to Find A^{-1} ?

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How to Find A^{-1} ?

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How to Find A^{-1} ?

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How to Find A^{-1} ?

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How to Find A^{-1} ?

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How to Find A^{-1} ?

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Inverse Method

4- Multiply $A^{-1}B$ to Find X.

$$\underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{X} = \underbrace{\begin{pmatrix} 3 & -1 & -1 \\ -4 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}}_{A^{-1}} \underbrace{\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}}_{B} = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$$

X = 2 **Y= -1** **Z = -2**

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Inverse Method

EXAMPLE

Consider the following system of linear equations:

1. Find the coefficient matrix A.
2. Find the inverse matrix of the coefficient matrix (A^{-1}).
3. Solve the system using the inverse matrix.

$$
\begin{aligned}
2x + 3y + z &= -1 \\
3x + 3y + z &= 2 \\
2x + 4y + z &= -2
\end{aligned}
$$

Inverse Method

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Inverse Method

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Inverse Method

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Check - Chapter Learning Outcomes

CLO

3

- Interpret the meaning of the solution of a system of linear equations.
- Express a system of linear equations in a matrix form.
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- Know some methods for solving a system of linear equations.
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- Find the inverse of a given matrix.
- Apply the inverse method for solving a system of linear equations.

End of Chapter

3